

Structural basis for androgen specificity and oestrogen synthesis in human aromatase

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Aromatase cytochrome P450 is the only enzyme in vertebrates known to catalyse the biosynthesis of all oestrogens from androgens^{1–3}. Aromatase inhibitors therefore constitute a front-line therapy for oestrogen-dependent breast cancer^{3,4}. In a three-step process, each step requiring 1 mol of O₂, 1 mol of NADPH, and coupling with its redox partner cytochrome P450 reductase, aromatase converts androstenedione, testosterone and 16 α -hydroxytestosterone to oestrone, 17 β -oestradiol and 17 β ,16 α -oestriol, respectively^{1–3}. The first two steps are C19-methyl hydroxylation steps, and the third involves the aromatization of the steroid A-ring, unique to aromatase. Whereas most P450s are not highly substrate selective, it is the hallmark androgenic specificity that sets aromatase apart. The structure of this enzyme of the endoplasmic reticulum membrane has remained unknown for decades, hindering elucidation of the biochemical mechanism. Here we present the crystal structure of human placental aromatase, the only natural mammalian, full-length P450 and P450 in hormone biosynthetic pathways to be crystallized so far. Unlike the active sites of many microsomal P450s that metabolize drugs and xenobiotics, aromatase has an androgen-specific cleft that binds the androstenedione molecule snugly. Hydrophobic and polar residues exquisitely complement the steroid backbone. The locations of catalytically important residues shed light on the reaction mechanism. The relative juxtaposition of the hydrophobic amino-terminal region and the opening to the catalytic cleft shows why membrane anchoring is necessary for the lipophilic substrates to gain access to the active site. The molecular basis for the enzyme's androgenic specificity and unique catalytic mechanism can be used for developing next-generation aromatase inhibitors.